

Exam Solutions

Advanced Methods in Applied Statistics

Author: Aske Roelshøj

UCPH-ID: KVF851

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I, Aske Roelshøj, expressively vow to uphold my scientific, academic, and moral integrity by working individually on this exam and soliciting no direct external help or assistance.

AI Declaration: I have used Github Copilot to assist with matplotlib formatting and dataframe LaTeX printing (see also last page).

Problem 1

1a+1b

We'll start by plotting the data, so we can get an idea of the type of functions might fit. The data from column 3 is plotted as a bar-plot, due to the discrete values.

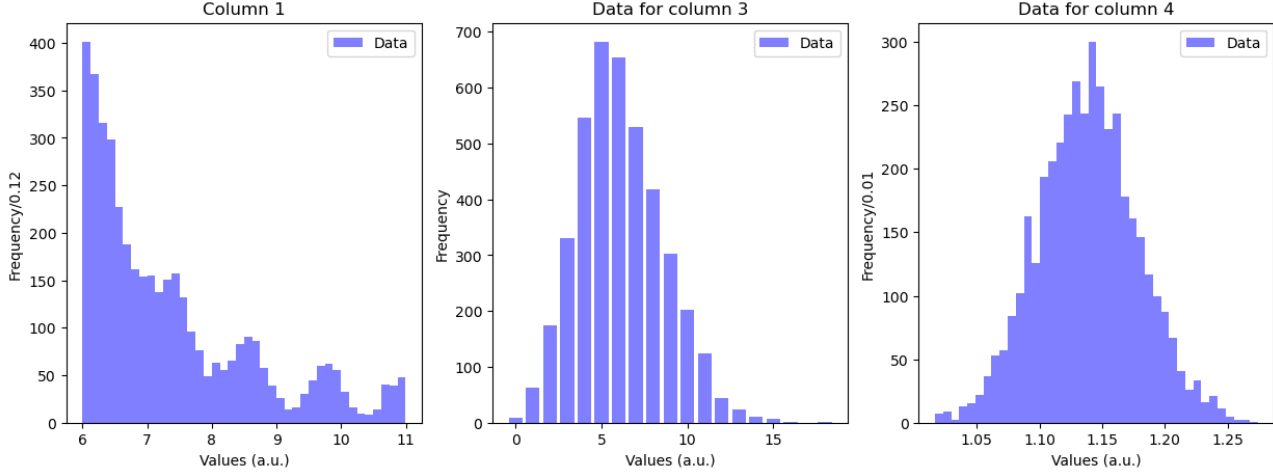


Figure 1: Data for column 1, 3 and 4 plotted together.

Looking at fig. 1, we might suspect that $f_8(x) = \sin(ax + d) + c \exp(bx) + 1$ fit the data from column 1, due to the in general exponentially decaying behaviour of the data with oscillations, adding d as a small tweak to the phase. The data from column 3 looks Poisson, $f_{11}(x = k)$, due to the discrete behaviour, and the data from column 4 looks Gaussian, $f_9(x)$. The Poisson distribution is assumed normalized, whereas we manually normalize both $f_8(x)$ and $f_9(x)$ using Scipy's quad.

Our approach for fitting is maximum likelihood using Minuit.

We'll start with column 1. We first do two 2d raster-scans, one of $a \in [0, 10]$ and $b \in [-10, 0]$, keeping $c = 5000$ and $d = 0.1$ fixed to get an idea for starting guesses, and one keeping a, b fixed with $c \in [4000, 8000]$ and $d \in [0.01, 0.5]$, and using the most likely values from the first raster-scan. All raster scans use 100 points in each direction. The results can be seen in Fig 2.

Using $a = 5.253$, $b = -1.010$, $c = 5939.394$ and $d = 0.426$ as starting guesses we do a maximum LLH fit to the data. We remember to limit all the parameters, such that they obey the limits given on exam problem, slide 7.

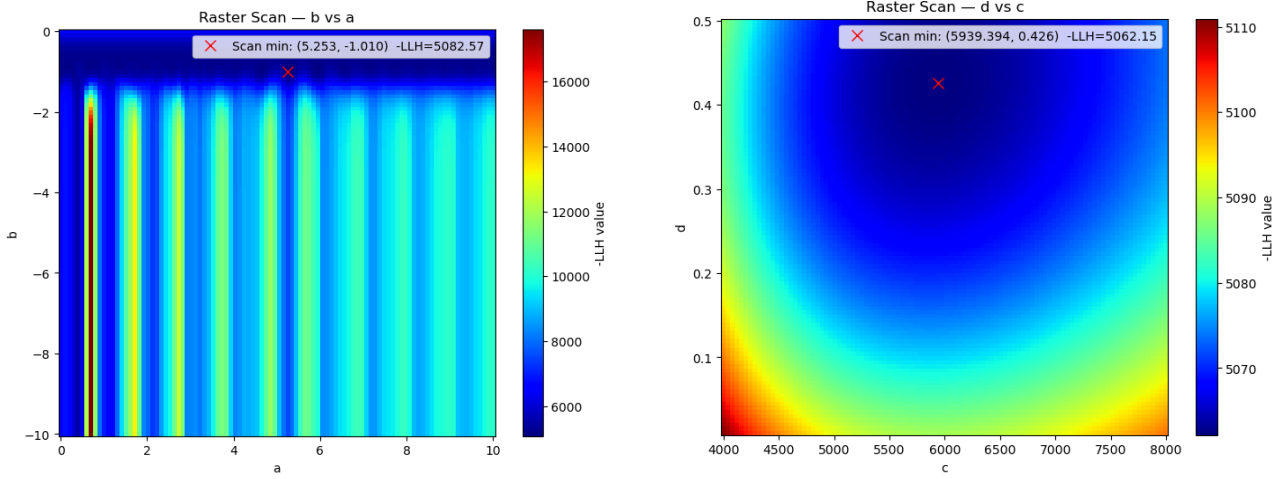
For data from column 3, we simply use the mean value of the data as the starting guess for $\lambda = 6.11$. And for the data from column 4 we simply use the mean and standard deviation from the data as starting guesses for the Gaussian, i.e. $a = 1.13$, $b = 0.04$.

The 1σ errors on the fit parameters are obtained from Minuit.

The best fits, with best fit values and errors are displayed on Fig. 3.

To argue for the best fits, we've done MC sampling for each column using the proposed functions and fitted parameters, using the minimum and maximum data-values as x-ranges. Then we do a two-sample KS test to see if the distributions agree. The results can be seen in Fig. 4.

The results are summarized in the following table Tab. 1.



(a) Raster scan: b vs a

(b) Raster scan: d vs c

Figure 2: Parameter scans showing $-LLH$ landscapes for (a, b) and (c, d) .

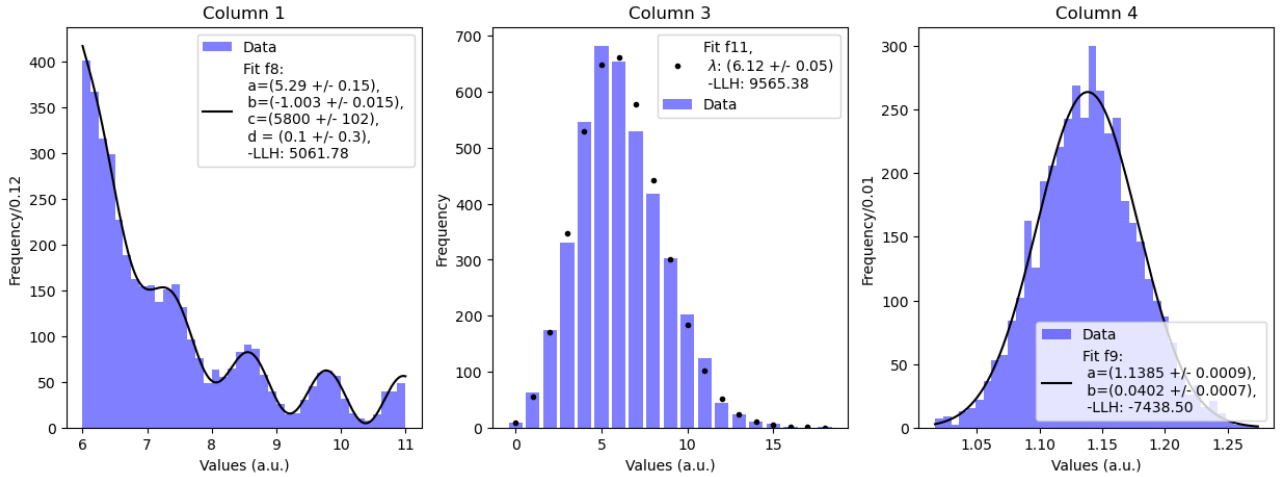


Figure 3: Data with best fit values and errors for column 1, 3 and 4.

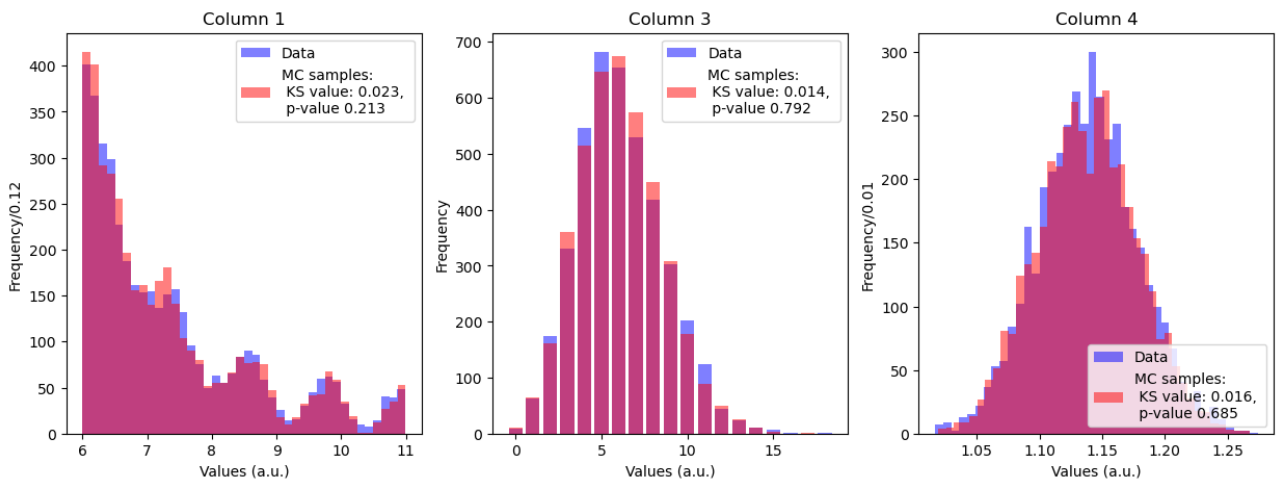


Figure 4: Data alongside MC sampling using the best fit values from Fig. 3, with KS and p-values.

As can be seen on Fig. 4, we obtain large p-values, indicating good agreement between the real data and the MC samples from the fitted functions, i.e. we cannot reject the null

	Column 1 Fit f8	Column 3 Fit f11	Column 4 Fit f9
a	5.29 ± 0.15	-	1.1385 ± 0.0009
b	-1.003 ± 0.015	-	0.0402 ± 0.0007
c	5800 ± 102	-	-
d	0.1 ± 0.3	-	-
λ	-	6.12 ± 0.05	-
KS statistic	0.023	0.014	0.016
p-value	0.213	0.792	0.685

Table 1: Table summarizing the results for parameters and errors for each model, with KS test statistic and p-value.

hypothesis that the distributions are the same.

All the results for this problem are summarized in Tab. 1

Problem 2

2a

We start by loading the data, and splitting into azimuth and zenith angles.

Then we create a MC-sampling algorithm that samples isotropic data, by uniformly sampling azimuth and zenith angles. We use seed 1. We start by sampling 139 points, i.e. the same amount as the data given, and plot these samples alongside the real data on a spherical area projection to visually verify the sampling algorithm, see Fig. 5.

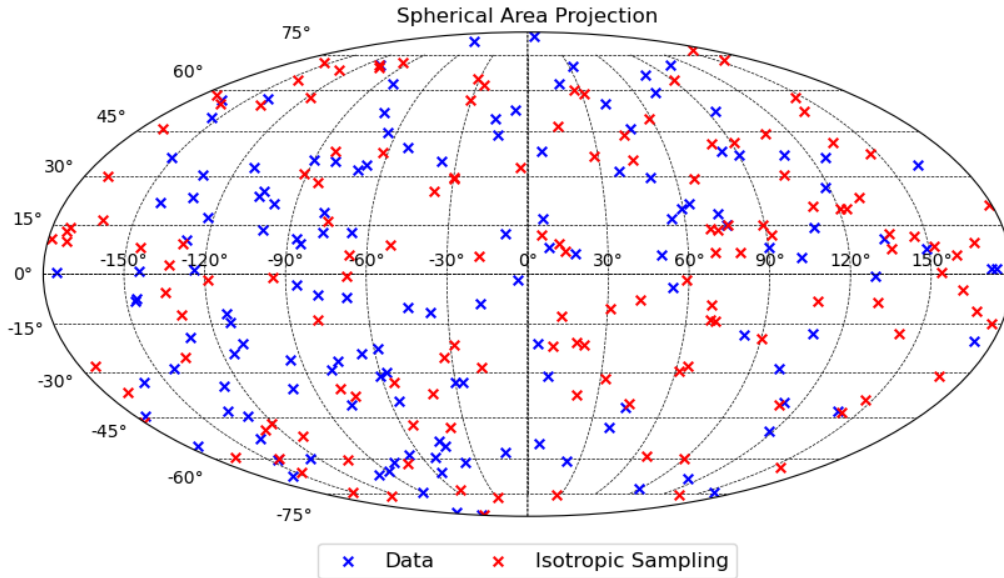


Figure 5: Plot of real and sampled isotropic data on a spherical area projection.

To quantify if the data is spherically isotropically distributed, we'll determine the two-point correlation function values for both real and isotropic sampled data. We'll sample $N_{point} = 139$ points $N_{exp} = 2500$ times, and create a KS test statistic distribution. Using the distribution we are able to determine the p-value associated with the KS value for the real data, and determine if it is spherically isotropically distributed.

The two-point correlation function, $\mathcal{C}(\{\mathbf{n}_i\}, \varphi)$, is defined as in lecture slides AhlersKosinen eq. (2) slide 31, with the predicted isotropic distribution defined as $\mathcal{C}_{iso}(\varphi) = 1/2(1 - \cos(\varphi))$ on slide 35. The KS test statistic is then defined as $KS = \sup_{\varphi} |\mathcal{C}(\{\mathbf{n}_i\}, \varphi) - \mathcal{C}_{iso}(\varphi)|$.

We transform our data to Cartesian coordinates, and determine the two-point correlation function and isotropic prediction for each point, for both real data and one set of samples. We plot the correlation function against $\cos(\varphi)$ on Fig. 6. As can be seen, the data bounces a bit away from the isotropic prediction, while the isotropic samples follow the prediction well.

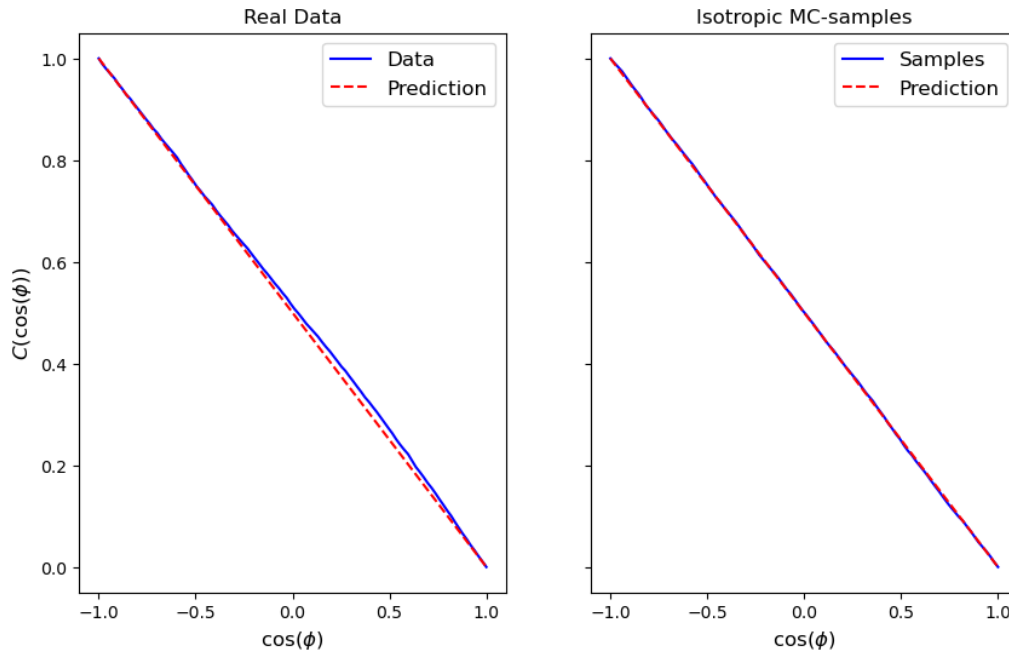


Figure 6: Two-point correlation function against isotropic prediction for real data and sampled data, both for 139 data points.

For the data on Fig. 6, we determine the KS test statistic to be $KS_{data} = 0.0221$ and $KS_{sample} = 0.0047$ for the real data and sampled isotropic data, respectively.

Sampling $N_{exp} = 2500$ times, we get the KS test statistic distribution shown on Fig. 7. On the figure, H_0 means the KS values associated with the MC-sampled isotropic distribution, i.e. it is the null hypothesis, while H_1 is the KS-value associated with our data.

We determine the p-value by taking the integral of KS-values for the isotropic samples for KS-values above KS_{data} . **We get a p-value of 0.0028.** With a 99%, i.e. 3σ confidence level we can conclude that the data is **not spherically isotropically distributed**.

2b

For this problem we adjust our sampling functions, such that we sample according to H_a and H_b . We then follow the same procedure as above, and get the results in Fig. 8.

We then calculate the p-values for both H_a, H_b w.r.t. H_1 , i.e. the real data. We get table 2.

From the table we get quite large p-values ($> 10\%$), larger than any reasonable significant level (3σ), and thus we can't reject that hypothesis A or B.

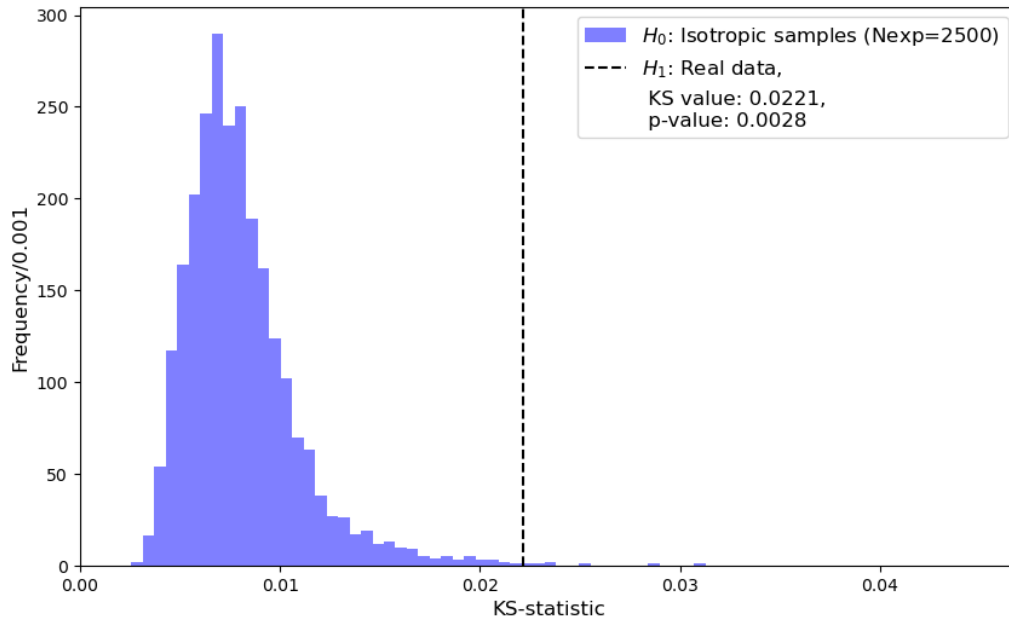


Figure 7: Distribution of KS values for 139 sampled isotropic values, 2500 times.

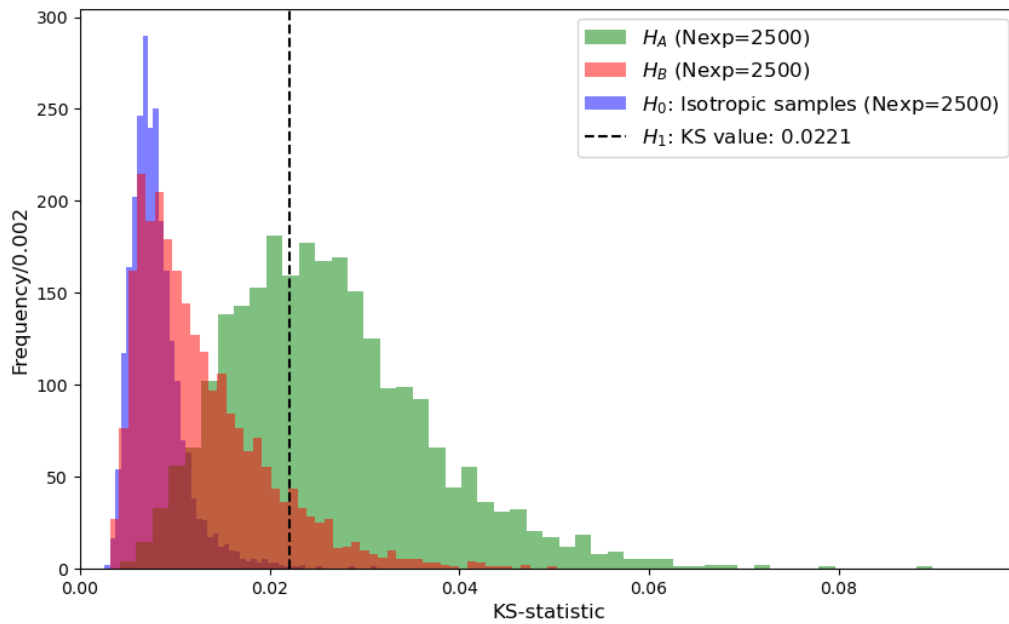


Figure 8: Distribution of KS values for H_0 (blue), H_a (green) and H_b (red) with $N_{exp} = 2500$.

Hypothesis	p-value vs H1
H_A	0.6176
H_B	0.1040

Table 2: P-values for H_a and H_b .

Problem 3

3a

We load and divide the data into distances and populations. To do a population estimate using a Gaussian KDE we use Scikit-Learn's KernelDensity function, with the Gaussian

kernel.

Setting a bandwidth of 25km, we fit on the distances using the population as sample weights, such that the kernels are relative to the size of population. The population density can then be found taking the log of the density, found using `.score_samples` and multiplying by the total population, here just the sum of the given population data. We do this for x-values from -100 to 1025 km away from Picton. The result can be seen in Fig. 9.

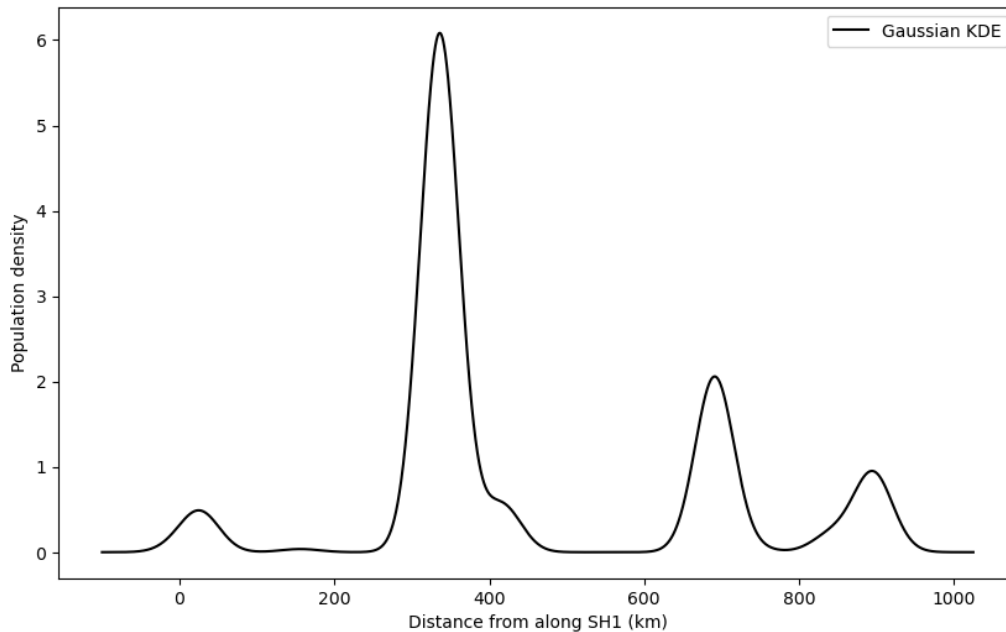


Figure 9: KDE using Gaussian kernel on population density as a function of distance from Picton station.

The total population estimate from 405 km to 430 km from Picton is then found using Numpys trapezoid, to calculate the integral between the boundaries. We get a population of 14365. Since the population data is given in thousands with one decimal, **we estimate a population of 14400.**

3b

We simply implement the boundaries and e-car densities, noting that Nelson-Marlborough is from 0 to 91 km away from Picton, since it is *along* SH1.

We then simply calculate the integrals between the boundaries (to get the amount of people) and multiply by the e-car densities. The results are seen in Tab. 3.

Region	Estimated E-Cars
Nelson-Marlborough	144
Canterbury	2426
Otago	803
Southland	131

Table 3: Estimated E-cars by region using e-car densities and Gaussian KDE from fig. 9.

3c

We will use the same approach as in 3a, but where we map the population in each city to the number of e-cars using the region-wise e-car densities. Since the e-car densities are region-based, we can simply calculate the number of e-cars by city by multiplying by the population in each city from the dataset by the region-wise e-car densities. The results can be seen in Tab. 4.

City	Region	Population	Estimated E-Cars
Picton	Nelson-Marlborough	4.3	23.65
Blenheim	Nelson-Marlborough	28.2	155.10
Kaikōura	Canterbury	2.3	13.34
Christchurch	Canterbury	381.0	2209.80
Ashburton	Canterbury	34.8	201.84
Dunedin	Otago	128.8	772.80
Milton	Otago	2.1	12.60
Gore	Southland	12.8	24.32
Invercargill	Southland	57.1	108.49
Bluff	Southland	1.8	3.42

Table 4: Calculated number of e-cars in each city based on population and region-wise e-car densities.

Using the number of e-cars in each city as weights, we can again use Scikit-Learn's `KernelDensity`. Here using the Epanechnikov kernel with bandwidth 35 km to obtain the prior. The results are in Fig. 10.

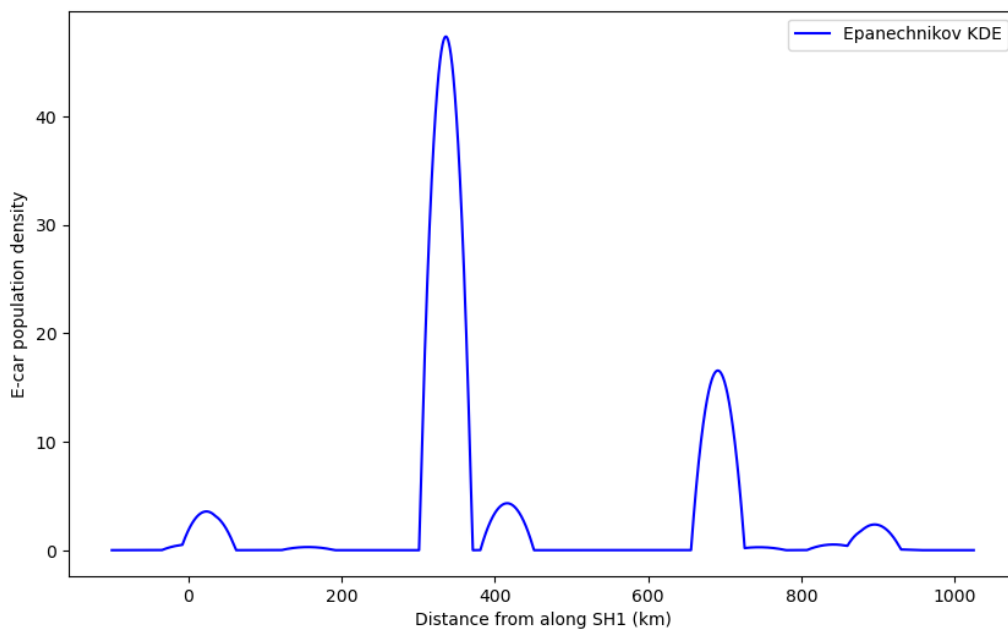


Figure 10: KDE using Epanechnikov kernel on e-car population density as a function of distance from Picton station.

We also load the camera positions and e-car count, such that we can use a Gaussian KDE with bandwidth 50km to determine the likelihood PDF. Here we use the e-car count from the cameras as weights. The results are shown in Fig. 11.

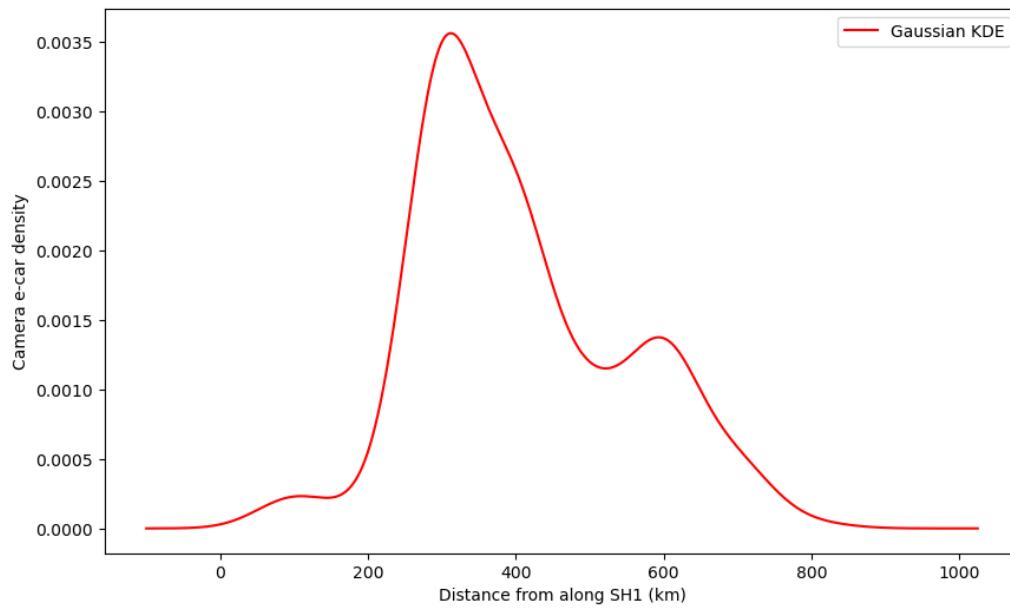


Figure 11: KDE using Gaussian kernel on e-car population density using based on cameras as a function of distance from Picton station.

We then multiply the likelihood and prior, and use trapezoid to normalize. The posterior, likelihood and prior are plotted in Fig. 12.

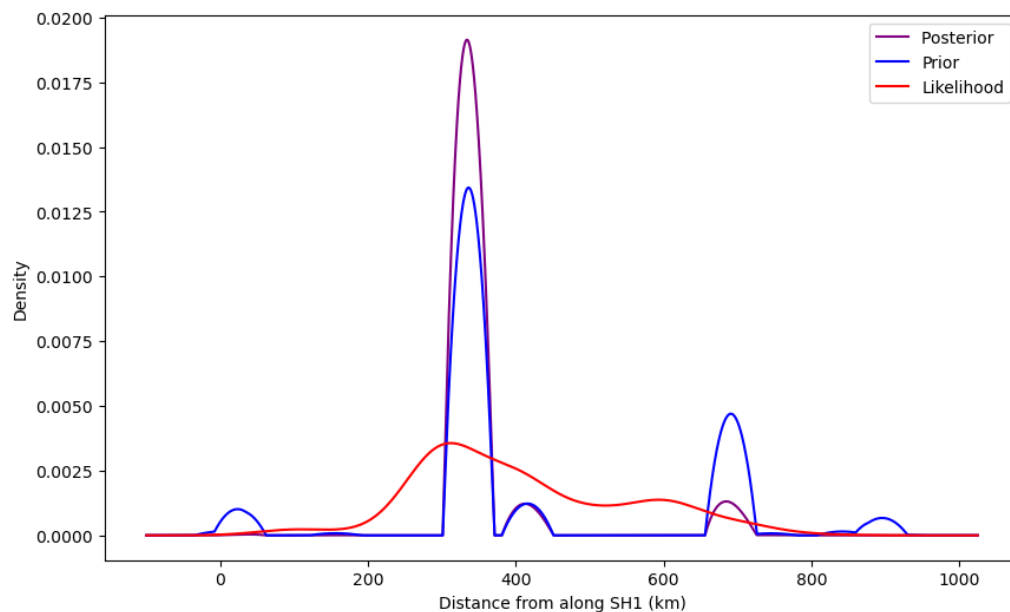


Figure 12: Posterior, Likelihood and Prior distributions as a function of distance from Picton.

We use our posterior PDF to determine the fraction of cars between 400km and 700km. **We obtain a fraction of 0.089, i.e. 8.9% of e-cars in New Zealand are between 400km and 700km along SH1.**

Problem 4

4a

We start by loading the data, and plotting it to get an overview on Fig. 13.

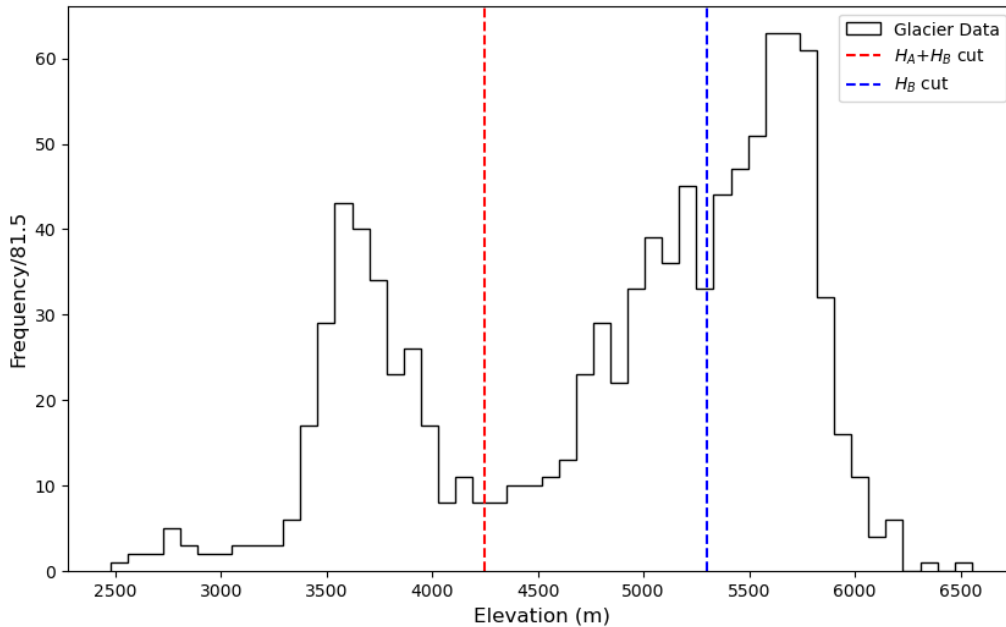


Figure 13: Histogram of the raw glacier data.

To fit the two hypothesis, we'll do a maximum LLH fit. For hypothesis A, we'll split the data in two by 4250 (red dashed), and for hypothesis B, we'll split the data in three by, 4250 and 5300 (blue dashed).

For the double gauss, we simply add two Gaussian functions with a scaling factor A , determining the share of the first Gauss (and $1-A$ for the second). For the triple gauss, we have 2 scaling factors, A_1, A_2 , again distributing the data-share that each Gaussian contributes with. As starting guesses, we use the means and standard deviations of the subset of data from each cut.

We obtain the results in Fig. 14 (next page).

As can be seen in Fig. 14, the triple Gaussian fit has a slightly lower -LLH value when compared to the double Gaussian fit, indicating a slightly better fit.

4b

To determine at what statistical significance hypothesis A can be accepted/reject when compared to hypothesis B, we'll use the likelihood-ratio, $-2\Delta LLH$, as a test statistic.

We get $-2\Delta LLH = 71.13$, and doing a χ^2 -test using $df = 3$, since the difference in parameters is 3, we get a p-value of $2.44 \cdot 10^{-15}$. We can thus with great confidence ($> 3\sigma$) reject hypothesis A in favour of B.

The p-value seems suspiciously small. One reason might be that the assumptions for Wilk's theorem might not be fulfilled. We do "only" have 1000 data points, and for the theorem we need "a large number of samples" (see AhlersKoskinen, slide 19) - 1000 data points might not be enough. Also the hypothesis might not be nested.

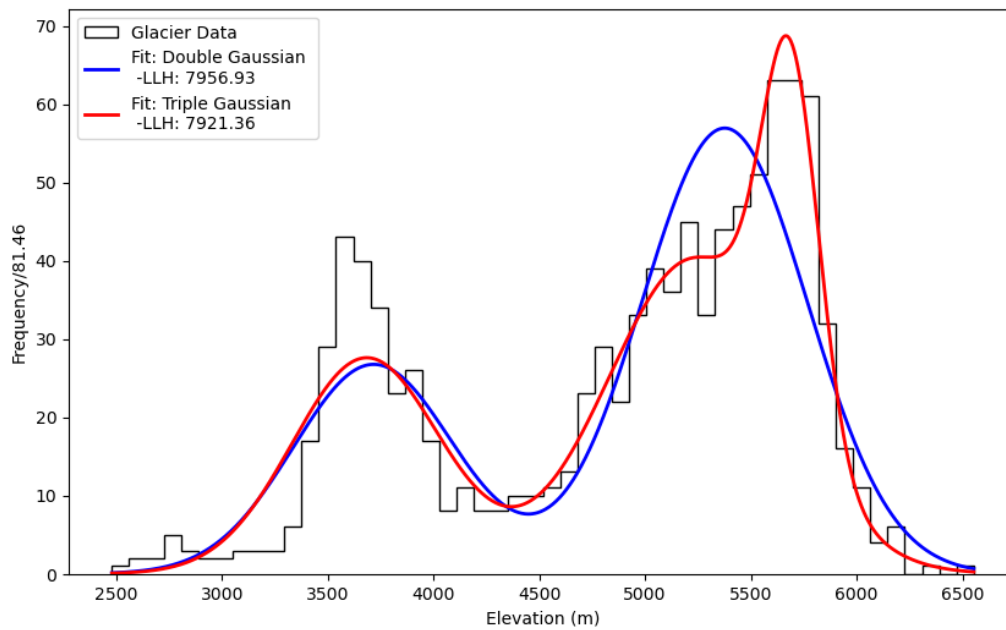


Figure 14: Double and triple Gaussian fits on glacier data.

Problem 5

5a

We'll use Numpy's `interp` to do the linear spline, and Scipy's `PchipInterpolator` to do the PCHIP spline.

The estimates for temperatures on sol 203.570 are summarized in table 5.

Method	$T(203.570)$ ($^{\circ}\text{C}$)
Linear	-115.913
PCHIP	-115.419

Table 5: Estimates for temperatures on sol 203.570 based on linear and PCHIP splines.

5b

We'll make 200 sol-values between 203.410 sol and 203.608 sol, and plot the results in Fig. 15.

If we expect the temperature to be monotonically dropping, one region to look at more closely would be between $[203.535, 203.558]$ sol (between green dashed lines in Fig. 15). Here we see what could look like a plateau or small increase in temperature, which is probably due to spline overcorrection.

5c

To investigate whether electronics with a temperature change tolerance of 0.09°C within 0.0004 sol should be replaced, we find the derivatives of the splines. We then find the steepest descent (maximum of the derivative of the spline), and multiply by 0.0004 to get the change of temperature in units of 0.0004 sols.

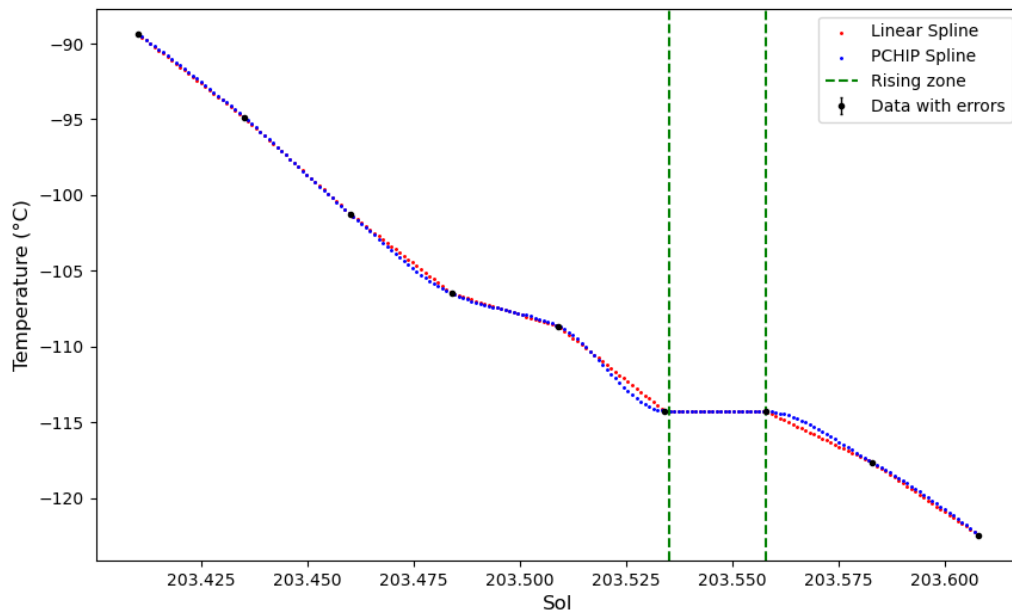


Figure 15: Scatterplot with real data, linear and PCHIP splines between 203.410 sol and 203.608 sol.

For the linear spline, we find a temperature change of 0.102°C within 0.0004 sols at 203.44 sols, and for the PCHIP spline a temperature change of 0.123°C within 0.0004 sols at 203.52 sols. The points are shown on fig. 16.

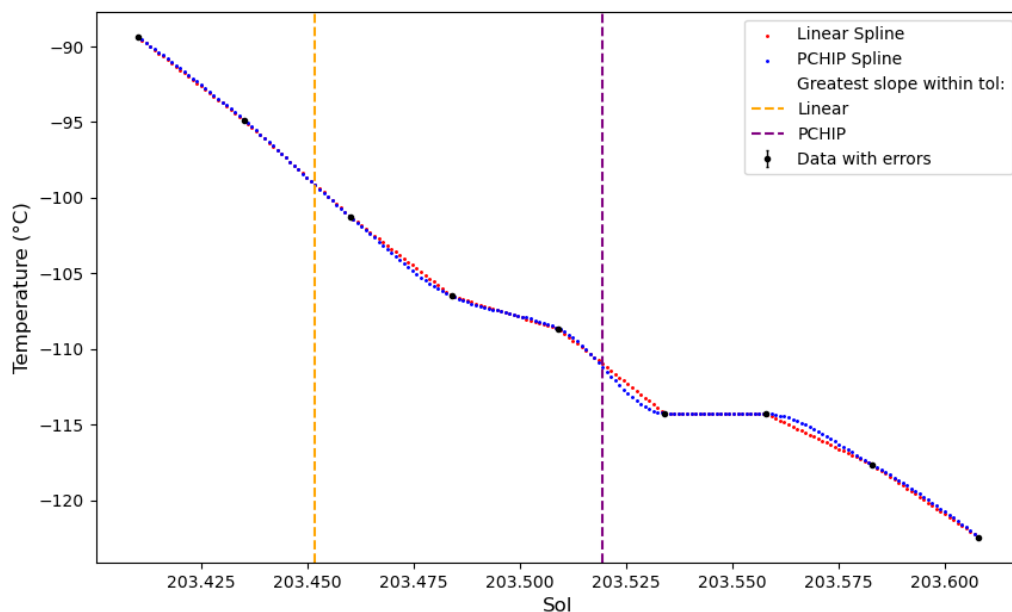


Figure 16: Scatter plot with real data, linear and PCHIP splines, with dashed lines indicating greatest slopes within step-tolerance of 0.004 sols.

Since we estimate temperature changes greater than 0.09°C within 0.0004 using both linear and PCHIP splines, **we conclude that new electronics will be needed.**

End of my solution.

UCPH's AI declaration

Declaration of using generative AI tools

☒ I/we have used generative AI as an aid/tool *(please tick)*

☐ I/we have **NOT** used generative AI as an aid/tool *(please tick)*

If generative AI is permitted in the exam, but you haven't used it in your exam paper, you just need to tick the box stating that you have not used GAI. You don't have to fill in the rest.

List which GAI tools you have used and include the link to the platform (if possible):

GitHub Copilot

Describe how generative AI has been used in the exam paper:

I have used Github Copilot to assist with matplotlib formatting and dataframe LaTeX printing.